

CHAPTER - 6 EVOLUTION

“The life that exists today is the result of a long process of evolution from simple to complex forms.”

1. INTRODUCTION

- **Evolution** is the gradual and irreversible change in heritable traits of a population over successive generations.
- It leads to the formation of new species from pre-existing ones.
- The process is driven by variations, natural selection and genetic drift.

Key Point

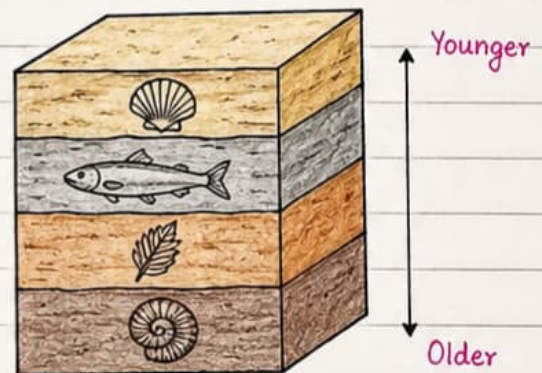
Evolution explains the diversity of life on Earth.

2. EVIDENCES OF EVOLUTION

(i) Palaeontological evidences (Fossils):

- Fossils are the preserved remains or impressions of ancient organisms.
- They show evolution of organisms over geological time.

Example → Archaeopteryx shows characters of both reptiles (teeth, tail) and birds (feathers, wings).



Fossil in Sedimentary Rocks

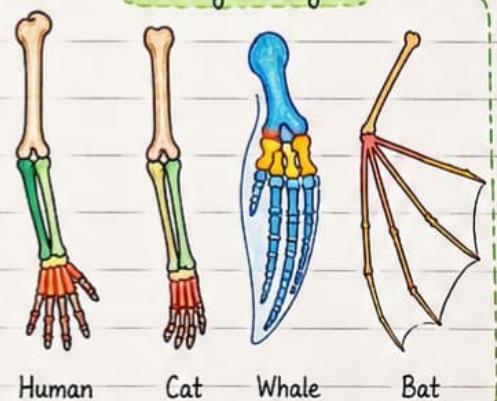
(ii) Comparative Anatomy:

- Similarities in the body structure of different organisms indicate common ancestry.
- Two types:
 1. **Homologous organs** – same basic structure, different functions.
 2. **Analogous organs** – different structure, similar functions.

Example →

Forelimbs of human, cat, whale and bat have same basic bones but different functions.

Homologous Organs



3. MECHANISMS OF EVOLUTION

- Evolution occurs due to changes in the genetic composition of populations over generations.

Key Mechanisms

(i) Variation

- Variations are differences in traits among individuals of the same species.
- They arise due to mutations and genetic recombination.

Example → Height, skin colour, blood group differences in humans.

Remember!

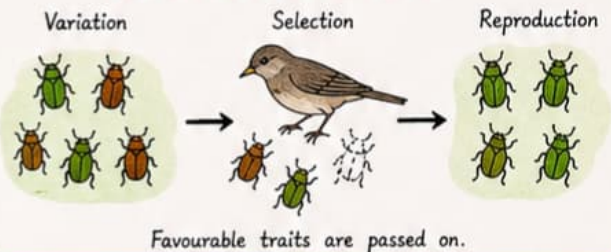
Variation is the raw material for evolution.

(ii) Natural Selection

- Individuals better adapted to the environment survive and reproduce more.
- Unfavourable traits are eliminated over time.

Example → Insects resistant to pesticides survive, while others die.

Natural Selection in Action



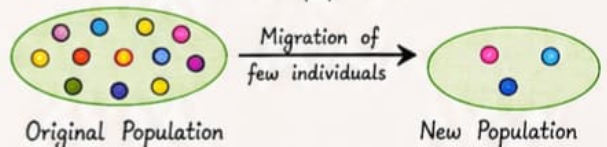
(iii) Genetic Drift

- Random changes in the frequency of genes in a small population.
- It is not due to natural selection.

Example → In small populations, some traits may disappear by chance.

Genetic Drift (Founder Effect)

A few individuals colonize a new area and form a new population.

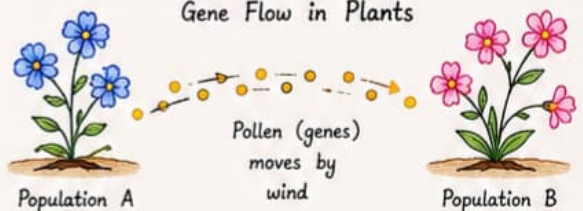


(iv) Gene Flow (Migration)

- Movement of individuals or gametes from one population to another.
- Introduces new genes and increases variation.

Example → Pollen transfer by wind from one plant population to another.

Gene Flow in Plants

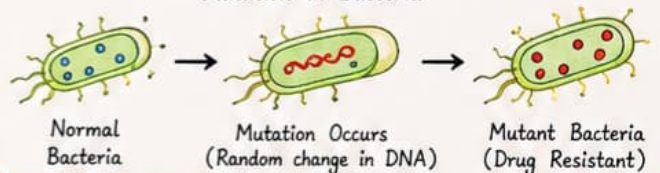


(v) Mutation

- Sudden and heritable changes in DNA sequence.
- Creates new alleles and new traits.

Example → Antibiotic resistance in bacteria due to gene mutation.

Mutation in Bacteria



In Short

Variation provides differences.
Mechanisms of evolution act on these variations.
Favourable traits are preserved; unfavourable are eliminated.
Over long periods → New species are formed.



CHAPTER - 6 : EVOLUTION

Date : / /

Page No. : 3

4. EVIDENCE FOR EVOLUTION

- Fossils, comparative anatomy, embryology, connecting links and molecular biology all provide strong evidence for evolution.

(i) Fossils

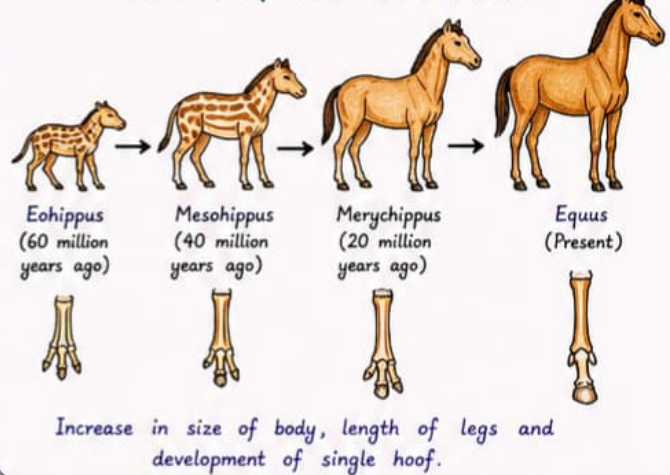
- Fossils are the remains, impressions or traces of ancient organisms preserved in rocks.
- They show the existence of organisms that lived in the past.
- Fossil record shows gradual change in the structure of organisms over time.

Example → Fossil of horse shows gradual change in size of body and legs.

Key Point

Fossils = History
Comparative Anatomy = Relationship
Embryology = Similar Origin
Molecular Evidence = Common Code

Evolution of Horse (Fossil Record)



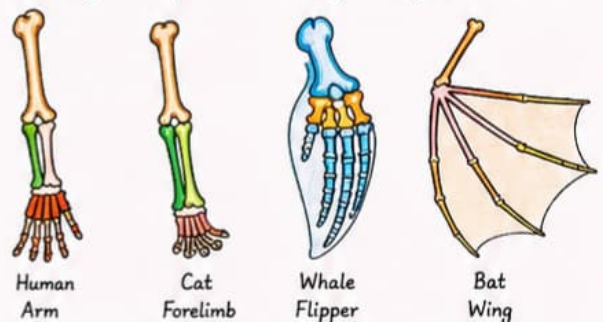
(ii) Comparative Anatomy

- Study of similarities and differences in the body structures of different organisms.

A. Homologous Organs – Same basic structure and origin, but different functions.

Example → Human arm, cat forelimb, whale flipper and bat wing have same structure but perform different functions.

Homologous Organs (Same Origin, Different Functions)



B. Analogous Organs – Different structure and origin, but similar functions.

Example → Wings of bird and wings of insect are used for flight but have different structure and origin.

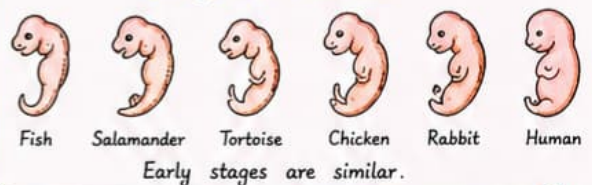
Analogous Organs (Different Origin, Similar Function)



(iii) Embryological Evidence

- Early embryos of different vertebrates are very similar in structure.
- This suggests that they have evolved from a common ancestor.

Embryos of Vertebrates



In Short



- ☆ Fossils → show past life and gradual change.
- ☆ Comparative Anatomy → shows similarities in structure.
- ☆ Embryology → shows similar origin.
- ☆ All these evidences support the theory of evolution.

Remember!

Similarities tell us about origin, differences tell us about evolution.

CHAPTER - 6 : EVOLUTION

Date : / /

Page No. : 4



5. EXAMPLES OF EVOLUTION

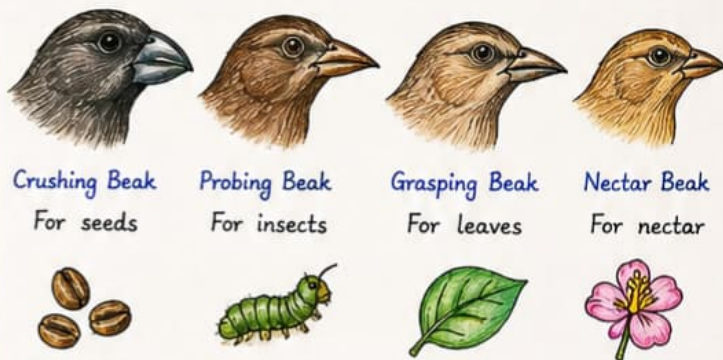
(i) Darwin's Finches

- On Galápagos Islands, finches with different beak shapes evolved.
- Each beak is suited for a specific type of food.
- This is an example of **adaptive radiation**.

Example → Finch with thick beak crushes seeds, while thin beak picks insects.



Darwin's Finches – Beak Variations



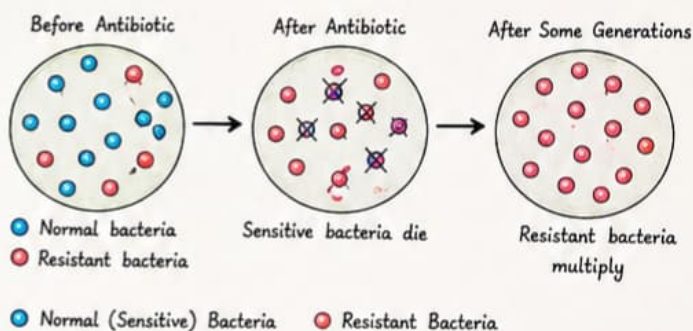
(ii) Antibiotic Resistance in Bacteria

- Random **mutations** produce some bacteria resistant to antibiotics.
- When antibiotics are used, sensitive bacteria die, resistant ones **survive**.
- Resistant bacteria multiply and pass on resistance to **next generation**.

Example → Overuse of antibiotics leads to drug-resistant strains like MRSA.



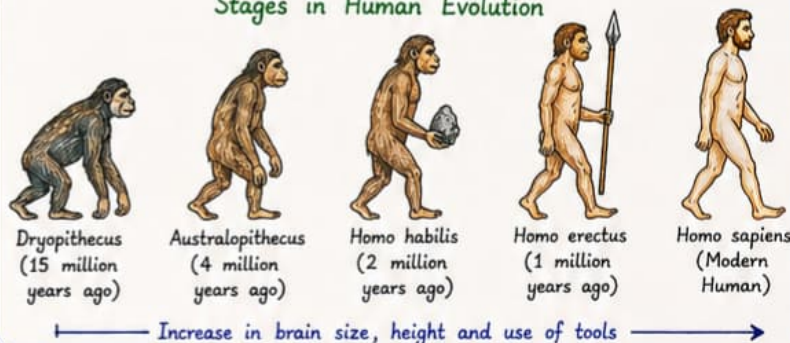
Antibiotic Resistance



(iii) Evolution in Humans

- Humans evolved from **ape-like ancestors** over millions of years.
- Changes occurred in **body structure, brain size, posture and tools**.
- These changes helped in **survival** and development of **civilisation**.

Stages in Human Evolution



6. SUMMARY - THEORY OF EVOLUTION



In Short

- ☆ Evolution is the gradual change in living beings over generations.
- ☆ It occurs through mechanisms like variation, natural selection and genetic drift.
- ☆ Fossils, comparative anatomy, embryology and molecular biology provide evidence.
- ☆ Evolution explains the **diversity of life** and the **origin of new species**.